Bounds Sharper Than Their Statements

A synthetic example

Lemma 1. *Let x, C and h be integers with $C \geq 0$, $h \geq 0$ and $x + 1 \leq Ch$. Then $x \leq Ch$.*

Theorem 2. *Let x, C and h be integers with $C \geq 0$ and $x + 1 \leq Ch$. If $h \geq 0$ then $x \leq 2Ch$.*